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CIS-PRJ-401 Research Project

DEVELOPMENT OF A BIOMETRIC AND MOTION-BASED IoT SECURITY SYSTEM FOR SMALL AND MEDIUM-SIZED ENTERPRISES: A CASE STUDY OF LIMBE, BLANTYRE

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ABSTRACT

This project presents the design and prototype development of a Biometric and Motion-Based IoT Security System for Small and Medium-Sized Enterprises (SMEs) in Blantyre, using Limbe SMEs as a case study. Many SMEs in Malawi rely on traditional security measures such as key-based access and manual surveillance, which are prone to key duplication, unauthorized access, and delayed incident response. To address this, the study developed a simplified, cost-effective IoT-based system that integrates fingerprint-based biometric authentication, PIR motion detection, automated alarms, and a centralized web-based monitoring dashboard.

The study adopted a pragmatism research philosophy with a mixed-methods approach, combining semi-structured interviews and direct observation for requirement gathering. The prototyping methodology was used for system development, allowing the design to be iteratively built, evaluated, and refined. The system architecture integrates an ESP32 microcontroller with fingerprint and PIR sensors at the hardware layer, a Next.js web application at the application layer, and PostgreSQL managed through Prisma and Neon together with Convex for real-time updates.

The developed prototype demonstrates that an affordable, integrated IoT security solution combining biometric access control and motion-based alerting is feasible for resource-constrained SME environments in Malawi, addressing a gap left by existing solutions such as ZKTeco, Bosch, and Hikvision, which are either too costly or too complex for SME adoption.

ACRONYMES AND ABBREVIATIONS

API	Application Programming Interface
DFD	Data Flow Diagram
ERD	Entity-Relationship Diagram
ESP32	Espressif 32-bit Microcontroller
FR	Functional Requirement
IoT	Internet of Things
MUBAS	Malawi University of Business and Applied Sciences
MVC	Model-View-Controller
MVP	Minimum Viable Product
PIR	Passive Infrared
NFR	Non-Functional Requirement
SME	Small and Medium-Sized Enterprise
SQL	Structured Query Language
UI	User Interface

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CHAPTER 1: INTRODUCTION

1.1. Background

Security in offices is essential for protecting business assets, sensitive information and ensuring employee safety (Kirkham, 2026). However, many organizations continue to rely on traditional methods such as physical guards, key-based access systems and closed-circuit television (CCTV) surveillance. These approaches can be fragmented, provide limited lack automation, and offer limited centralized monitoring. In Malawi, many small and medium-sized enterprises (SMEs) face challenges in adopting modern security systems due to high costs and limited access to integrated technologies. As a result, some SMEs continue to rely on manual or standalone systems that may contribute to unauthorized access and delayed responses to security incidents.

The emergence of the Internet of Things (IoT) has enabled interconnected devices to communicate and exchange data, supporting automation and real-time monitoring. IoT-based systems can integrate technologies such as biometric authentication, motion detection and alarm systems to monitor environments and respond to security threats (Kumar et al., 2019). Biometric authentication provides secure access control using unique characteristics of individuals (Tripathi & Nishad, 2020), while motion detection can identify movement within monitored areas and trigger predefined responses such as alarms (ASE, 2024). When these technologies are integrated with a centralized web-based dashboard, security events can be monitored and managed from a single platform.

This study therefore focuses on developing an IoT-based security system that combines fingerprint authentication, PIR motion detection, automated alarms, and centralized web-based monitoring. The system uses an ESP32 microcontroller to connect the physical security components with the web-based platform, providing an integrated approach to security management suitable for SME environments.

1.2. Problem Statement

Observations from local business environments indicate that many SMEs in Malawi continue to rely on traditional security systems that are associated with security risks such as key duplication, unauthorized access and delayed response to security incidents. Although advanced security solutions exist, they are often expensive, complex, and require infrastructure, beyond the capability of many SMEs.

The reliance on traditional security methods introduces several vulnerabilities. Key duplication can allow unauthorized individuals to access restricted areas without proper verification, putting assets and sensitive information at risk. Unauthorized access increases the likelihood of theft, data breaches, and misuse of resources, especially where user identity cannot be properly verified. In addition, delayed response to security incidents means threats are often detected only after damage has occurred, leading to financial losses, operational disruption, and reduced employee safety. These issues weaken overall security and can threaten the stability and sustainability of SMEs.

Therefore, there is a need for a simplified and cost-effective IoT-based security system that integrates biometric authentication, motion detection, automated alarms, and a centralized web-based monitoring dashboard to improve access control, security event detection, and response for SMEs in Malawi.

1.3. Aim of the Study

The main aim of this study was to develop a simplified and cost-effective IoT-based smart security system for SMEs in Malawi that integrates biometric authentication, motion detection, automated alarms, and a centralized web-based monitoring dashboard.

1.4. Objectives of the Study

The study was guided by the following specific objectives:

1. To identify security requirements of Small and Medium-Sized Enterprises in Blantyre through semi-structured interviews and direct observation.

2. To develop a hardware prototype comprising an ESP32 microcontroller, an AS608 fingerprint sensor, a PIR motion detection sensor and a buzzer alarm.

3. To develop a web-based monitoring dashboard incorporating fingerprint user management, real-time access logging, motion event detection and device status monitoring.

4. To integrate the hardware prototype with the web-based dashboard through a RESTful HTTP API to form a unified IoT security system.

5. To evaluate the developed prototype against defined performance parameters including fingerprint authentication accuracy, motion detection response time, false alarm rate and system reliability.

1.5. Research Questions

1.5.1. Main Research Question

How can a cost-effective IoT-based smart security system integrating biometric authentication, motion detection, and centralized monitoring be developed for SMEs in Blantyre?

1.5.2. Specific Research Questions

1. What security requirements and challenges do SMEs in Blantyre experience that can be addressed using IoT-based security system?

2. How can biometric authentication, motion detection and a centralized dashboard be effectively designed and integrated into a secure, unified and cost-effective IoT-based system for SMEs?

3. How can a functional prototype be developed to integrate the hardware components with the web-based monitoring platform?

4. How does the total cost of the developed prototype compare with the purchase cost of similar products?

1.6. Significance of the Study

The study is significant to SMEs because it provides a practical approach to improving office security through the integration of biometric authentication, motion detection, automated alarms, and centralized monitoring. The proposed system is designed with affordability and simplicity in mind, making it more suitable for small and medium-sized business environments.

The study also contributes to the application of IoT technologies in security systems within the Malawian context. It demonstrates how low-cost hardware such as the ESP32 can be integrated with web-based software to provide centralized monitoring and security event management. This can be used in fields such as Farming or Irrigation in agriculture to effectively monitor and manage farm activities.

Furthermore, the project provides a foundation for future improvements in IoT-based security systems, including the integration of additional sensors, improved alert mechanisms, and support for multiple security devices.

1.7. Scope of the Study

This study focused on the development of a prototype IoT-based smart security system for SMEs in Blantyre. The system integrates fingerprint-based biometric authentication, PIR-based motion detection, automated alarms, an ESP32 microcontroller and a centralized web-based monitoring dashboard.

The software component includes a web-based interface for managing authorized users, monitoring security events, viewing device status, and managing security alerts. The system uses PostgreSQL with Prisma and Convex services to support data management and real-time system operations.

The study was limited to a prototype implementation and did not include advanced features such as network cybersecurity, fire detection, or large-scale enterprise security infrastructure.

CHAPTER 2: LITERATURE REVIEW

2.1. Introduction

This chapter reviews literature related to IoT-based security systems, biometric authentication, motion detection, and centralized security monitoring. It discusses key concepts relevant to the study, examines existing security systems and their limitations, identifies the research gap, and presents the development framework guiding the proposed system.

2.2. Definition of key terms

In order to understand the concepts that were be used in this study, it is important to define the key terms related to the proposed system.

a. **Internet of Things (IoT)**

The Internet of Things (IoT) refers to a network of interconnected physical devices that collect, exchange, and process data through communication networks. IoT enables sensors, microcontrollers, and software applications to work together to support automation and real-time monitoring (Kumar et al., 2019).

In this study, IoT enables communication between the ESP32 microcontroller, security sensors and the web-based monitoring dashboard. This allows security events such as fingerprint authentication and motion detection to be transmitted to the centralized system for monitoring and response.

b. **Biometric Authentication**

Biometric authentication is a security mechanism that verifies an individual's identity based on unique physical or behavioral characteristics such as fingerprints or facial

features (Tripathi & Nishad, 2020). Unlike traditional keys or passwords, biometric authentication uses characteristics that are directly associated with an individual, making it suitable for access control applications.

This study uses fingerprint authentication because fingerprints provide a practical balance between identification accuracy, speed, cost, and ease of implementation. The AS608 fingerprint sensor is used to capture and match fingerprints at the device level before the authentication result is communicated to the backend system.

c. **Motion Detection**

Motion detection involves identifying physical movement within a monitored environment and initiating a predefined response. Sensors such as Passive Infrared (PIR), ultrasonic, and microwave sensors can be used for motion detection (ASE, 2024). PIR sensors detect changes in infrared radiation produced by objects such as humans and can be used to identify movement within a monitored are

In this study, PIR technology is used to detect movement within the office environment, particularly when the premises are not expected to be occupied. Detected motion can result in an alarm event and notification through the monitoring dashboard.

d. **Motion Detection**

Centralized security monitoring refers to the management and observation of security events through a single platform. Instead of relying on separate devices or standalone systems, security information can be presented through a centralized dashboard.

In this study, the web-based dashboard provides administrators with a central interface for monitoring security events, viewing device status, managing authorized users, and responding to alerts. This improves visibility of security activities and reduces dependence on manually checking individual devices.

2.3. Overview of IoT-Based Security System

Traditional office security approaches commonly depend on physical guards, mechanical keys, standalone alarms, or surveillance systems. Although these approaches can provide basic protection, they may have limitations in areas such as automation, centralized monitoring, and immediate response to security events.

IoT-based security systems address some of these limitations by connecting sensors, processing devices, communication networks, and software platforms. Security events can be detected by sensors, processed by microcontrollers, transmitted through a network, and presented to users through a centralized application.

For example, a biometric sensor can authenticate an individual attempting to access a secured area, while a motion sensor can detect movement when the premises are unattended. When integrated with an IoT platform, these events can be reported and notified to a security administrator.

The proposed system applies this approach by integrating an ESP32 microcontroller with fingerprint and PIR sensors, an automated alarm, and a web-based monitoring platform. This provides both access control and environmental monitoring within a single system.

2.4. Discussions of similar systems and their limitations

Several IoT-based security systems have been developed to improve organizational security. However, many of these systems present limitations when applied to SMEs.

a. **ZKTeco Biometric Access Control System**

ZKTeco provides biometric access control solutions that support technologies such as fingerprint authentication for controlling access to restricted areas (ZKTeco, 2026).

These systems provide reliable identity verification and are suitable for organizations requiring dedicated access control.

However, many ZKTeco solutions are primarily focused on access control and may require additional systems or integration to provide broader functionality such as PIR-based motion detection and customized centralized monitoring. This can limit flexibility for organizations seeking a single low-cost integrated solution.

b. **Bosch Building Integration System**

Bosh offers solutions that integrate intrusion detection, surveillance and alarms systems through a centralized platform (Bosch, 2026). Such systems provide extensive security capabilities and are appropriate for environments requiring advanced security infrastructure.

However, the level of infrastructure, configuration, and finance associated with such systems can make them less friendly for smaller organizations with limited financial and technical resources.

c. **Hikvision security systems**

Hikvision provides security technologies including surveillance, access control and intelligent security monitoring with trained AI models (Hikvision, 2026). Its solutions can support remote monitoring, quick security tips and suggestions, and integration of multiple security components.

Despite these capabilities, the configuration, maintenance, and infrastructure requirements of comprehensive commercial security solutions can present challenges for SMEs. Smaller organizations may therefore require simpler systems that focus on essential security functions while remaining affordable and easier to maintain.

In the context of Malawi, although there is limited documented research on IoT-based security systems, organizations such as TNM, Airtel Malawi and the National Bank of Malawi have employed advanced biometric and surveillance systems. However, these are complex and costly, making adoption extremely challenging for SMEs.

2.5. Research Gap

The reviewed systems demonstrate that biometric authentication, motion detection, surveillance, and centralized monitoring can be used to improve organizational security. However, there remains a gap in providing a simplified and affordable integrated solution specifically suited to the conditions of SMEs in Malawi.

ZKTeco provides effective biometric access control but does not necessarily provide the customized integration of biometric authentication, motion detection, automated alarms, and a centralized web dashboard required by this study. Bosch provides extensive integrated security capabilities but can involve infrastructure and complexity that may exceed the requirements of smaller organizations. Hikvision provides comprehensive security and surveillance solutions, but similar concerns regarding cost, configuration, and maintenance may affect adoption by SMEs.

Therefore, the gap addressed by this study is the development of a simplified IoT-based security system that combines fingerprint authentication, PIR motion detection, automated alarms, and centralized web-based monitoring within a single platform. The system is designed around affordable and readily available components such as the ESP32, while using web technologies to provide centralized monitoring and management.

2.6. Development Framework

The development of the proposed system is guided by a layered IoT architecture consisting of four conceptual layers: the perception layer, network layer, processing layer, and application layer (Maurya et al., 2024).

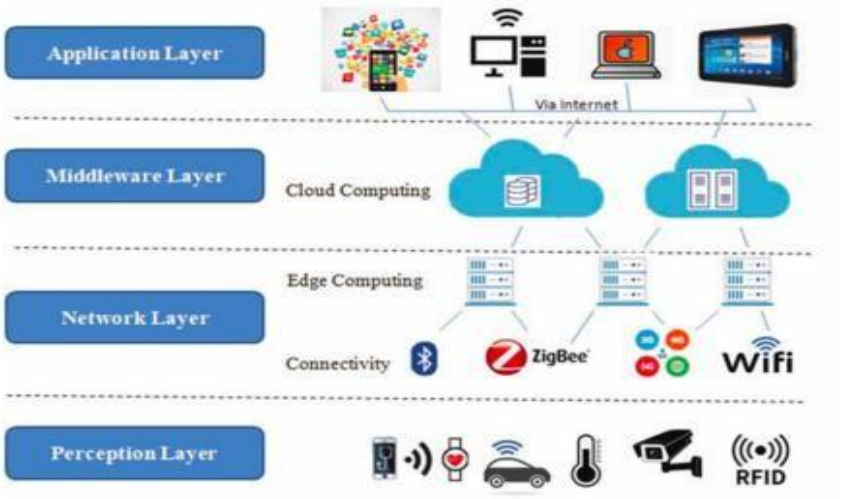


Fig 2.1 Layered IoT Development Framework

a. Perception Layer

The perception layer consists of the physical devices responsible for collecting information from the environment. In this study, the layer includes the ESP32 microcontroller, AS608 fingerprint sensor, PIR motion sensor, LEDs, and buzzer.

b. Network Layer

The network layer facilitates communication between the physical devices and the software system. The ESP32 uses Wi-Fi to communicate with the web application through HTTP-based API requests. This enables security events and device information to be transmitted to the backend.

c. Processing Layer

The processing layer manages the storage, processing, and delivery of security information. The proposed system uses a Next.js application together with PostgreSQL accessed through Prisma and Convex services. PostgreSQL is used for relational application data, while Convex supports real-time security event data and reactive updates.

d. Application Layer

The application layer provides services directly to the administrator. In this study, it is represented by the web-based dashboard developed using Next.js. The dashboard allows administrators to manage authorized users, monitor security events, view device status, and respond to security alerts.

The layered framework provides a logical structure for integrating the hardware and software components of the proposed system. It also supports the system's main objective of providing centralized and real-time security monitoring.

2.7. Conclusion

This chapter reviewed the major concepts and technologies related to the proposed IoT-based security system. It discussed IoT, biometric authentication, motion detection, and centralized security monitoring, followed by a review of existing systems and their limitations. The review identified a gap in affordable and integrated security solutions suitable for SMEs in Malawi. The chapter also presented the layered IoT development framework that guides the integration of the system's hardware, communication, processing, and application components.

CHAPTER 3: METHODOLOGY

3.1. Introduction

This chapter describes the methodology used to conduct the study and develop the proposed IoT-based security system. It presents the research philosophy, research approach, system development methodology, sampling method, data collection methods, data analysis procedures, ethical considerations, and limitations of the study.

3.2. Research Philosophy

The study adopted a pragmatism research philosophy, which focuses on implementing practical solutions and real-world applications. Pragmatism was appropriate because the study aimed to investigate a real-world security problem and develop a practical technological solution for SMEs in Blantyre (Jansen, 2023).

This philosophy supported the integration of both qualitative and quantitative methods, necessary for requirement gathering, system development and performance evaluation (Elgeddawy & Abouraia, 2024). Qualitative methods were used to understand the security challenges and requirements of SMEs, while quantitative measures were used to evaluate the performance of the developed prototype. This combination provided both contextual understanding of the problem and measurable evidence of the system's performance.

3.3. Research Approach

This study adopted a mixed research approach, combining both qualitative and quantitative methods. Qualitative approach was used to gather in-depth understanding of existing security functionalities, user needs and aspirations, and challenges faced by SMEs through interviews and observations. Quantitative approach was applied during

system evaluation, measuring performance metrics such as fingerprint recognition accuracy, motion detection response time, false alarm rates and dashboard alert latency (Shorten & Smith, 2017). The combination of these approaches provided a broader understanding of the security problem while also providing measurable evidence of the effectiveness of the developed system.

3.4. System Development Methodology

The system adopted the Prototyping Model methodology. The prototyping approach was selected because it allowed the proposed security system to be developed incrementary, tested and evaluated, and refined based in identified requirements and observations during development (Liyanage, 2025). both the hardware and software components needed to be integrated and tested before producing the final prototype.

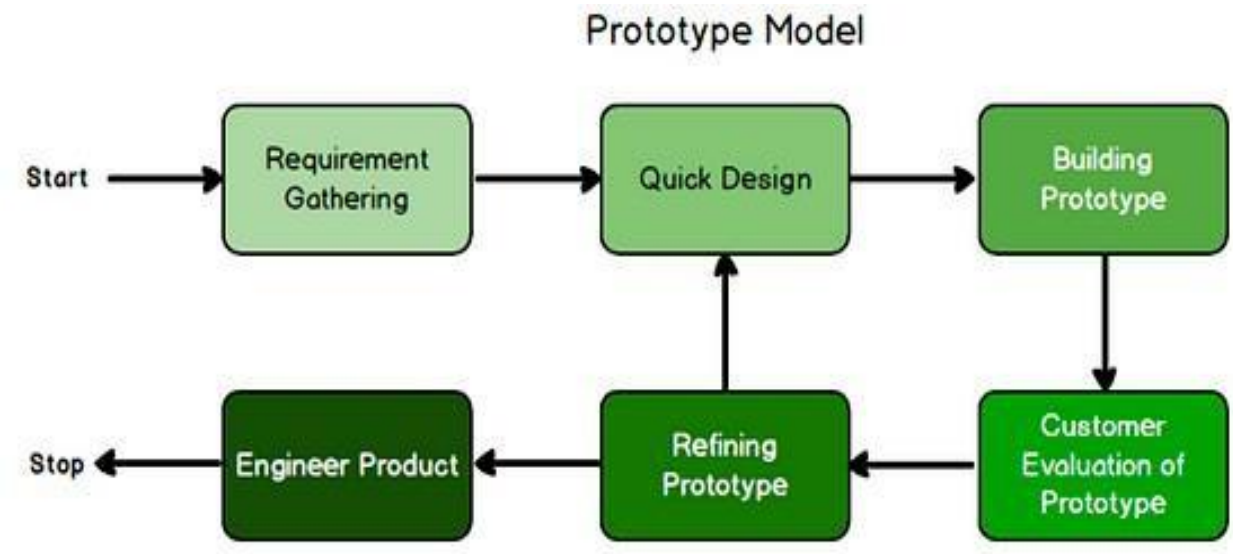


Fig 3.1: Prototyping Systems Development Lifecycle Stages

The development process consisted of the following stages:

Stage	**Description**
Requirement Gathering	Security requirements were identified through interviews, observations, and analysis of existing security practices.
Quick Design	An initial design of the hardware components, system architecture, database structure, and user interface was prepared based on the identified requirements.
Prototype Development	The ESP32, AS608 fingerprint sensor, PIR motion sensor, buzzer, and other hardware components were integrated with the Next.js web application and backend services to develop a prototype.
Prototype Evaluation	The prototype was tested to determine whether the implemented functions met the identified requirements and to identify technical issues.
Refinement	Identified issues were corrected and the hardware, software, and system processes were refined based on the test results.

Final System Development	The refined components were integrated into the final working prototype and subjected to further testing and evaluation.

3.5. Sampling

The study used **purposive sampling** to select participants from SMEs in Blantyre. This sampling technique was appropriate because participants were selected based on their relevance to the study and their experience with office security practices and business operations (EBSCO, 2026). Since this study is more qualitative in nature, a sample size formula is not required. Instead, the sample is determined by the need to collect relevant information rather than statistical representation.

SMEs were selected for the study. From each SME, two owners or managers and four employees were targeted, giving a total of **10 owners or managers and 20 employees**. Participants were selected based on their involvement in security-related activities, business operations, or use of office facilities. Participants were also required to have at least one year of experience within their respective organizations.

The selected participants provided information about existing security practices, challenges, and requirements that informed the design and evaluation of the proposed system.

3.6. Data Collection

The study used several data collection methods to obtain information required for the analysis, development, and evaluation of the proposed system.

The following were the data collection methods that were used:

1. **Semi-structured Interviews**

Semi-structured interviews were conducted with selected participants to gather information about existing security practices, challenges, and requirements. The semi-structured format allowed the researcher to use prepared questions while providing participants with an opportunity to provide additional information relevant to the study.

2. **Direct Observation**

Direct observation was used to examine security practices and the physical environments of the selected SMEs. This helped identify potential security challenges and understand how security-related activities were performed in the workplace.

3. **System Logs**

System records and logs generated during prototype testing were also reviewed to evaluate system activities such as authentication attempts, motion detection events, and alarm events. These records provided supporting information for evaluating the operation of the developed system.

During prototype testing, biometric authentication and motion detection data were collected from consenting participants under controlled conditions. The collected data was used to evaluate authentication performance, motion detection, response time, and system reliability.

3.7. Data Analysis

The data collected during the study was analysed using both qualitative and quantitative techniques. Qualitative data obtained from interviews and direct observations was analysed thematically. The responses and observations were reviewed to identify common themes related to existing security practices, challenges, system requirements, and user expectations.

Quantitative data obtained during prototype testing was analysed using performance measures. These included fingerprint authentication accuracy, motion detection response time, false alarm occurrences, and system reliability. The results were used to determine whether the developed prototype met the identified functional requirements and to identify areas requiring improvement.

The findings from both forms of analysis were used to guide system development and evaluate the effectiveness of the proposed IoT-based security system.

3.8. Ethical Considerations

Ethical considerations were observed throughout the study to protect the rights, privacy, and confidentiality of the participants. Participants were informed about the purpose of the study and the activities involved before taking part. Participation was voluntary, and informed consent was obtained from participants before interviews, observations, and prototype testing were conducted.

Participant information was treated confidentially and was only used for purposes related to the study. Personal information was not unnecessarily disclosed in the report or system demonstrations. During biometric testing, participants' fingerprint information was handled securely and was only collected with their consent. Where applicable, test data was removed after completion of the study.

Permission was obtained before conducting observations or collecting information within the selected SMEs. The researcher also ensured that the study activities did not unnecessarily interfere with normal business operations or expose participants to avoidable risks.

3.8. Limitations of the Study

The study had several limitations, these include:

a. **Limited Testing Environment:** Testing was conducted in a controlled environment, which may not fully represent real-world office conditions such as

network instability or environmental variations. To mitigate this, testing scenarios will be designed to simulate common real-world conditions as closely as possible.

b. **Dependence on Internet Connectivity:** Since the system relies on IoT and a web-based dashboard, its performance is dependent on stable internet connectivity. Network failures may affect real-time monitoring and communication between devices. This can be mitigated by implementing local data caching on the ESP32 to retain access attempts during connectivity interruptions.

c. **Hardware Constraints:** The use of low-cost components such as ESP32 and PIR sensors may limit performance and detection accuracy. These constraints are acknowledged as trade-offs necessary to maintain affordability for SMEs, and can be addressed in future iterations by upgrading to more capable hardware.

CHAPTER 4: REQUIREMENTS ANALYSIS

4.1. Introduction

This chapter presents the analysis of the requirements for the proposed biometric and motion-based IoT office security system. It describes the functional and non-functional requirements, the system-s primary users and the business processes modeled through data flow diagrams and use case diagrams. The models define what the system will do.

4.2. Functional Requirements

Below are the functional requirements of the proposed system:

Table 4.1: Functional Requirements Table

Requirement ID	Functional Requirement
FR1	The system will allow registered administrators to successfully log into the web dashboard.
FR2	The system will register and manage authorized users together with their fingerprint IDs.
FR3	The system will authenticate users using fingerprint recognition.
FR4	The system will deny access to unauthorized fingerprint authentication attempts.

FR5	The system shall detect motion during non-working hours using PIR sensors and notify the administrator through the dashboard.
FR6	The system will display real-time alerts on the dashboard.
FR7	The system will allow the administrator to manage authorized users.
FR8	The system will display the statuses of connected and active ESP32 devices.
FR9	The system will allow the administrator to acknowledge and resolve alarms.

4.3. Non-Functional Requirements

Below are the non-functional requirements for the proposed system:

Table 4.2: Non-Functional Requirements Table

Requirement Type	Description
Security	The system will only allow authorized administrators to access the dashboard.
Performance	The system should respond to authentication and motion events with within 5 seconds.

Reliability	The system will ensure high availability with minimal downtime.
Usability	The dashboard will offer a simple and friendly user interface, making it easy to navigate.
Scalability	The system shall support more IoT-devices in the future.
Maintainability	The software shall be modular to accommodate updates or maintenance.
Availability	The system will be accessible anywhere while the webserver is running.

4.4. System Users

The proposed system is designed to support the following primary users:

Administrator

The administrator is responsible for managing the overall security system through the web-based dashboard. They can register and manage authorized users, monitor access attempts and motion detection events, view security alerts, manage connected IoT devices, and view system logs. The administrator also responds to security incidents by taking appropriate action whenever unauthorized access or motion is detected.

Authorized User

An Authorized user is an individual whose fingerprint has been enrolled in the system. When requesting access, the users fingerprint is scanned and verified against a stored

record of biometric vectors. If authentication is successful, access is granted; otherwise, access is denied and the event is recorded in the system.

Unauthorized User

An unauthorized user is an individual who's not permitted to access the system. Their biometric fingerprints are not registered in the system and they may pose as a security threat. If they attempt to utilize biometric access, the system rejects access to them and notifies the administrator through the dashboard.

4.5. Business Process Analysis

Now, to understand the system's operational flow and interactions, the business processes were modeled using standard analysis techniques, including data flow diagrams and use case diagrams on "*draw.io"*.

4.5.1. Context Diagram (Level 0 DFD)

The context diagram, or Level 0 DFD, illustrates the system as a single process, defining its boundary and its data exchanges with external entities.

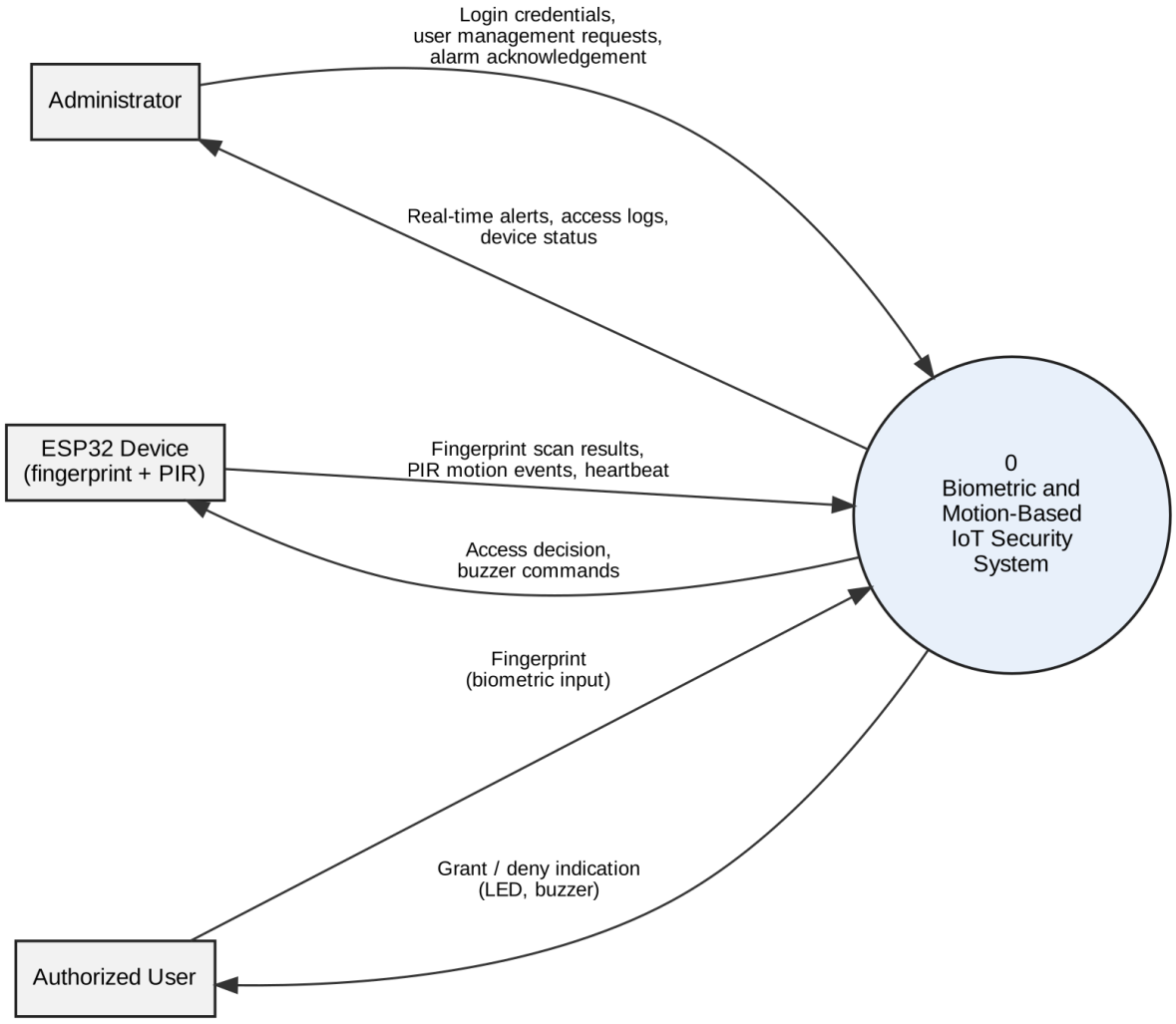


Figure 4.1 presents the Level 0 (context) DFD for the system.

The Administrator authenticates and issues management requests, receiving real-time alerts and dashboard data in return. The ESP32 device reports fingerprint scans and PIR motion events, and receives access-grant/alarm commands. The Authorized User interacts with the system indirectly by presenting a fingerprint at the device, receiving an implicit grant/deny result via the device's LED indicators and buzzer.

4.5.2. Context Level 1

The Level 1 DFD (Figure 4.2) decomposes the Level 0 DFD system into four main sub processes: Enrollment, Auth check, Motion Alerts and Monitoring. This illustrates the primary functions and data flows between the processes.

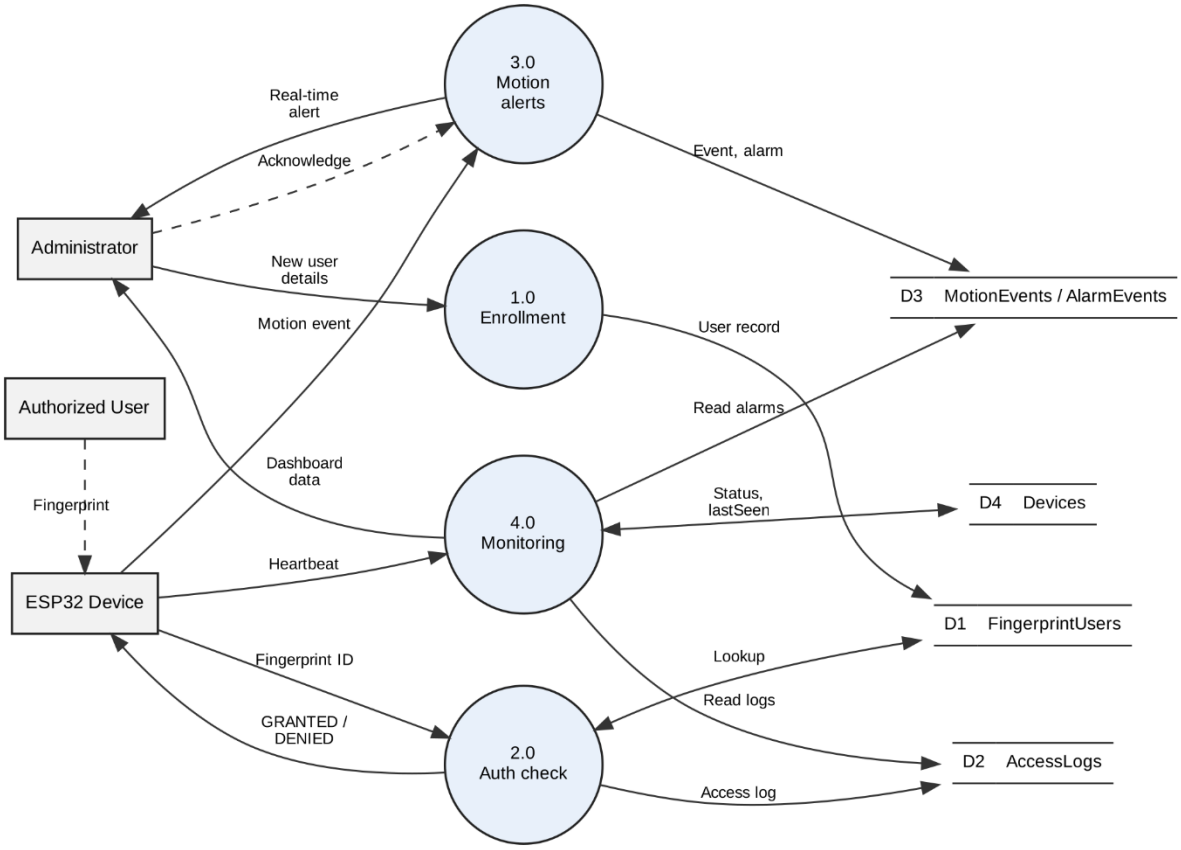


Figure 4.2 presents the Level 1 DFD for the system.

4.6. Use cases

The use case diagram illustrates the functionalities available to each actor identified in section 4.4, capturing their goals and how they interact with the system.

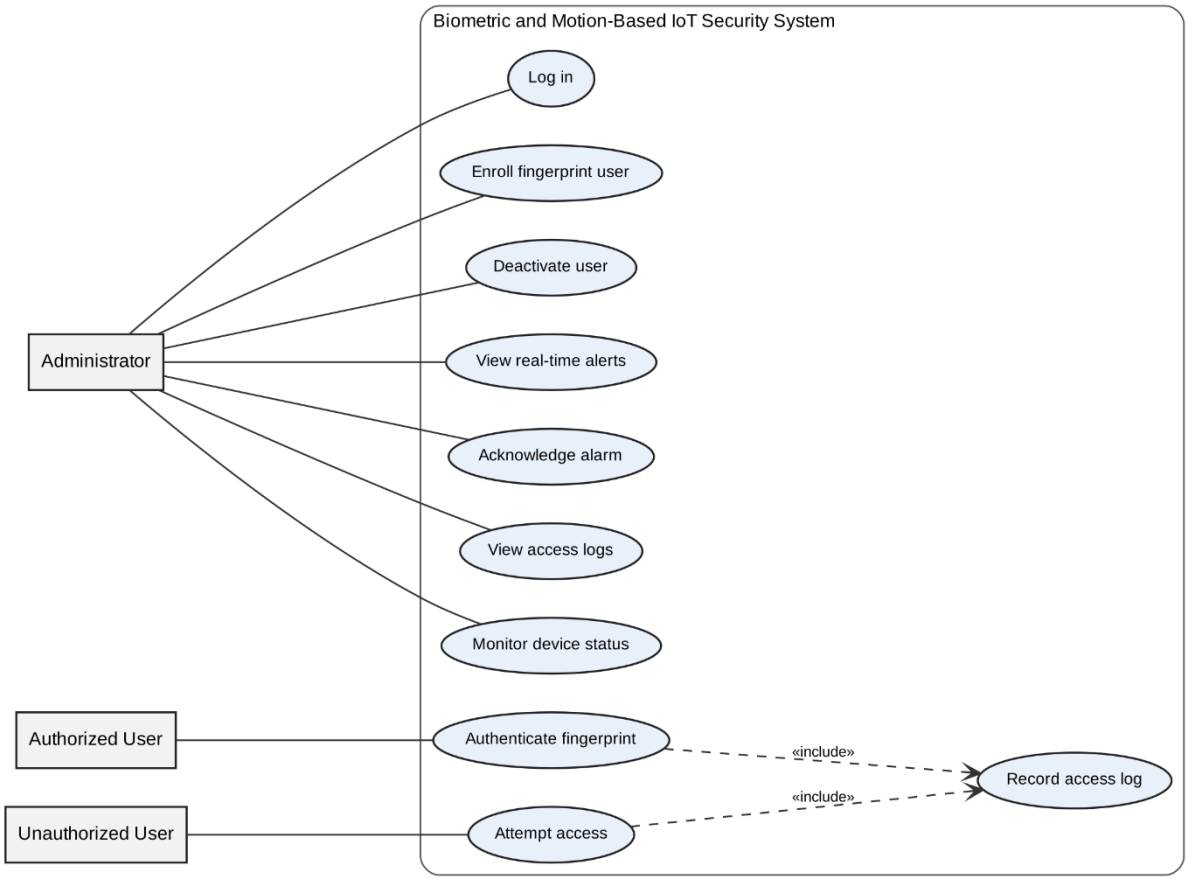


Figure 4.3 shows the use case diagram for the system.

The Administrator is the primary actor for all dashboard-facing functionality which include: logging into the system, enrolling authorized users' fingerprints, viewing real-time security alerts, and monitoring connected ESP32 device status. The Authorized User interacts with the system exclusively through the Authenticate use case, presenting a fingerprint at an enrolled device to request access. This separation reflects the distinct roles defined in Section 4.4, where the Administrator manages the system, while the Authorized User is a subject of it.

CHAPTER 5: SYSTEM DESIGN

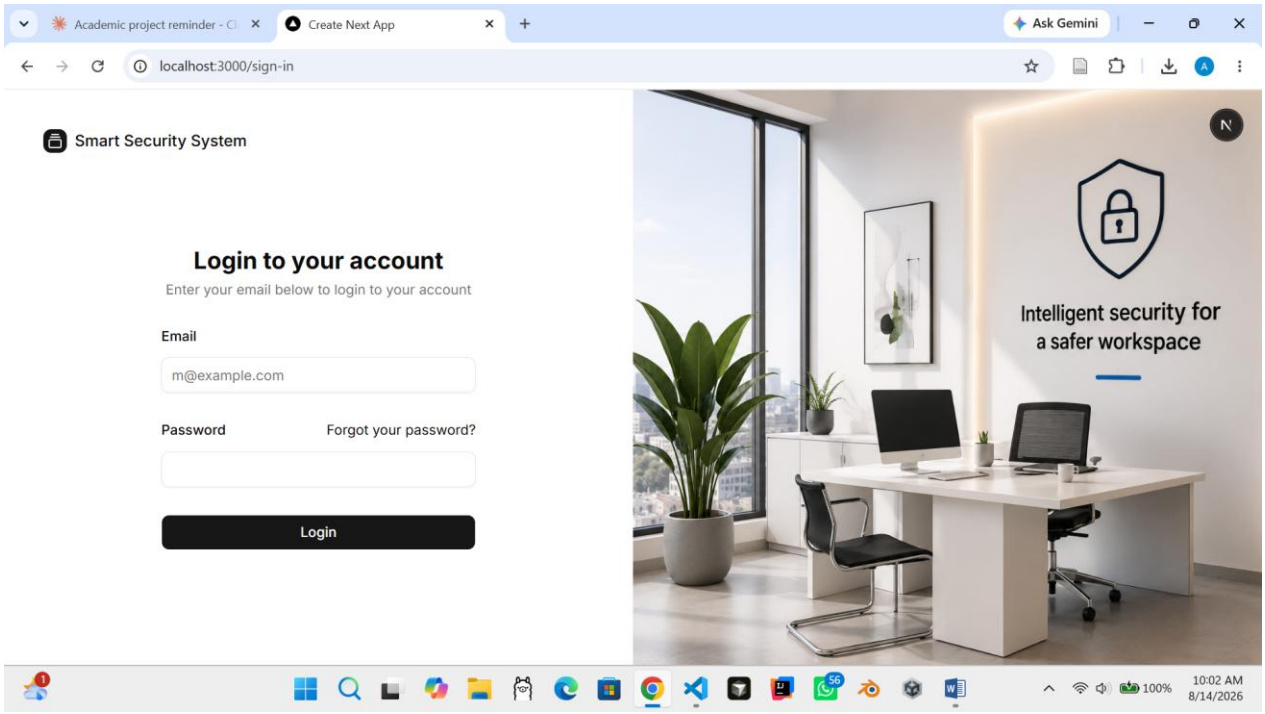
5.1. Introduction

This chapter details the technical design of the biometric and motion-based IoT security system, translating the requirements identified in chapter 4 into a concrete blueprint for implementation. It covers the interface design, data design, process and algorithm design, and the system architecture.

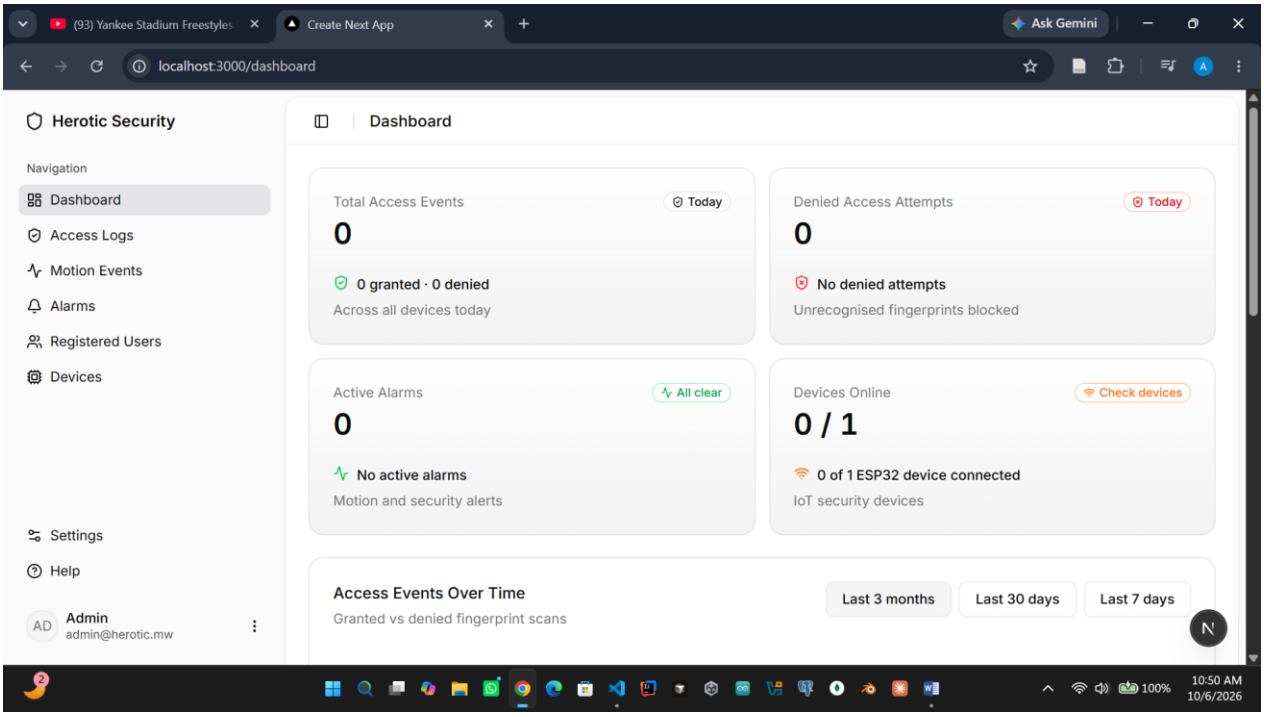
5.2. Interface Design

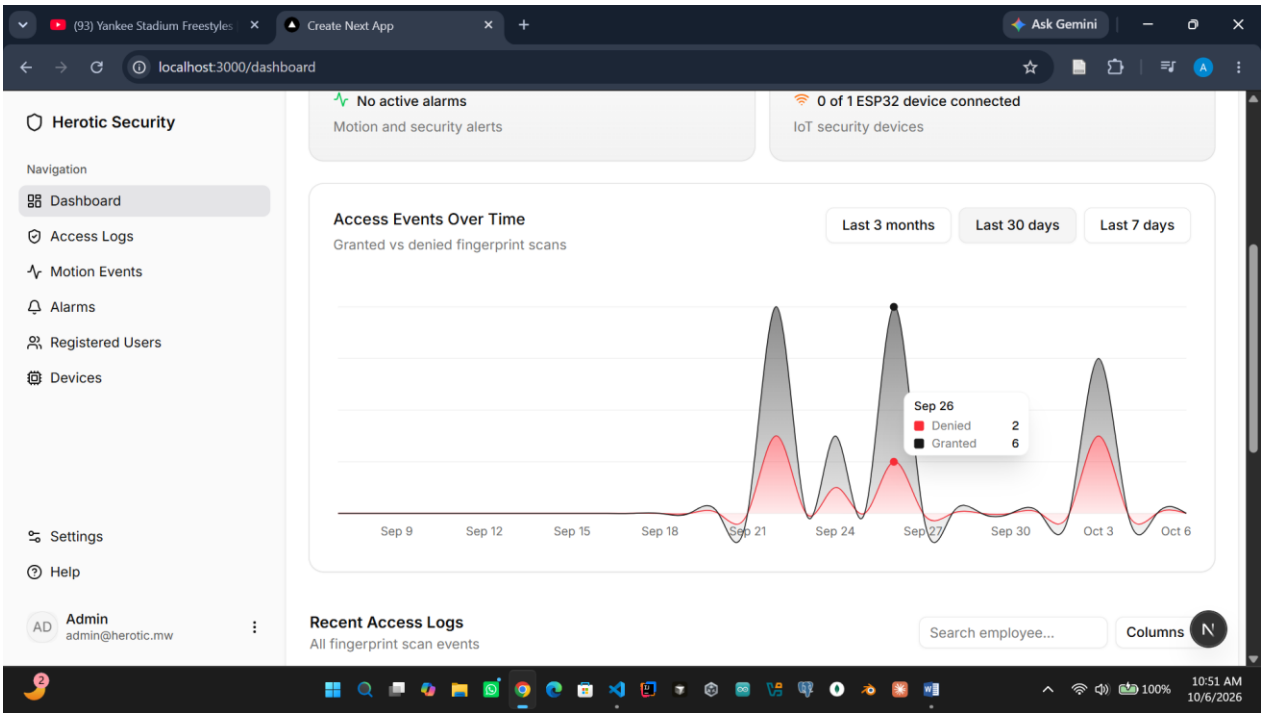
The systems web-based dashboard is designed to give the Administrator a clear, secure and centralized view of the security system.

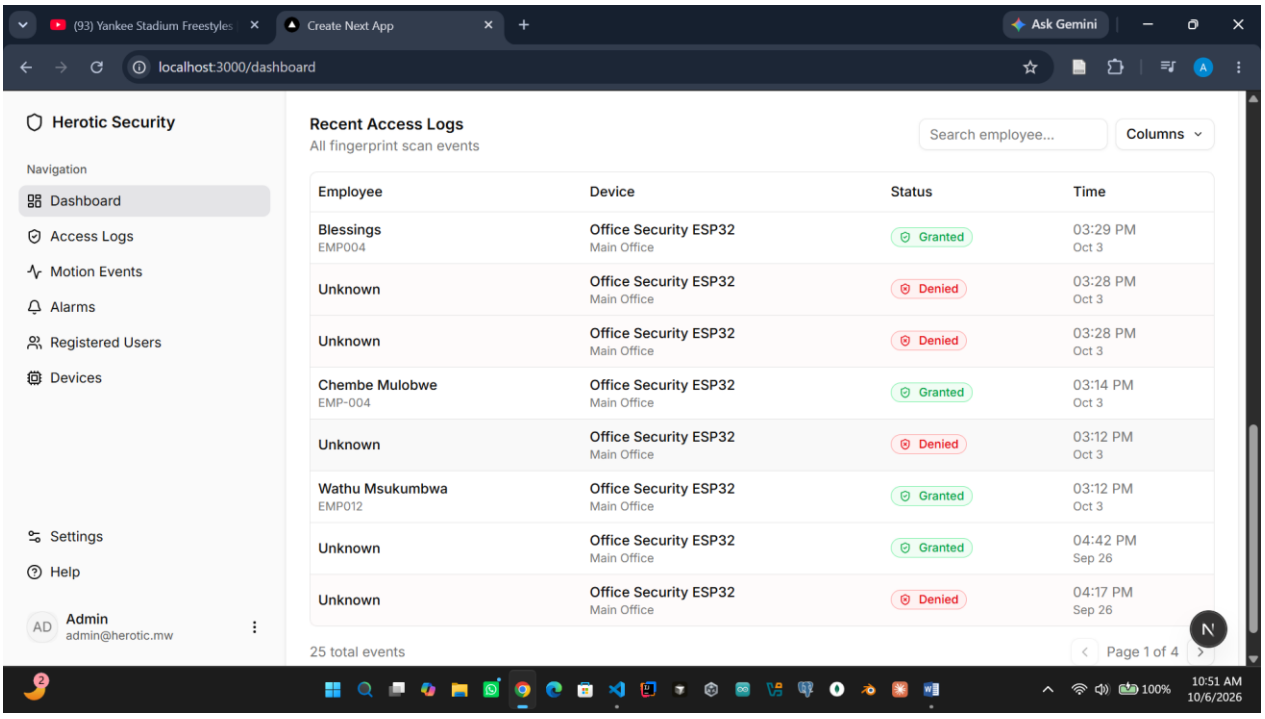
1. Login Page: A secure entry point requiring administrator credentials before any dashboard content is accessible.



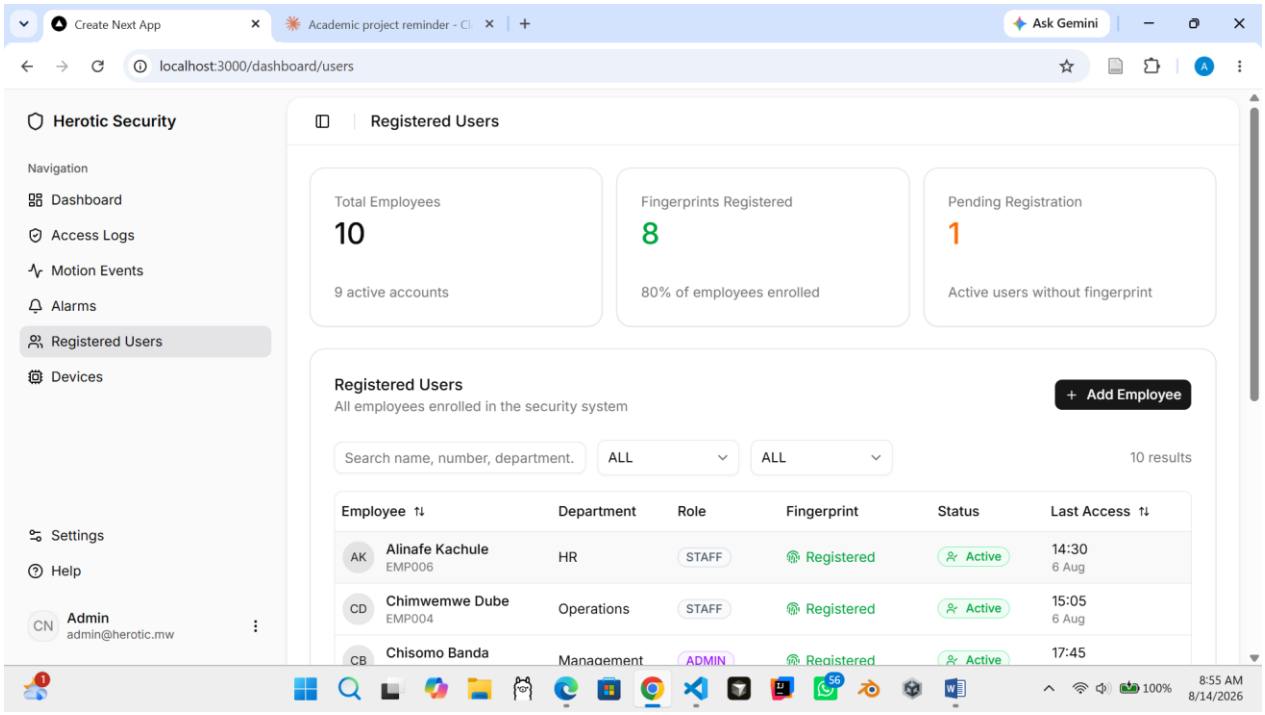
2. Main Dashboard: Page that displays connected devices status, a real-time feed or recent alerts and a summary of key security statistics. It also includes a small audit log of biometric activities that occurred during the day.



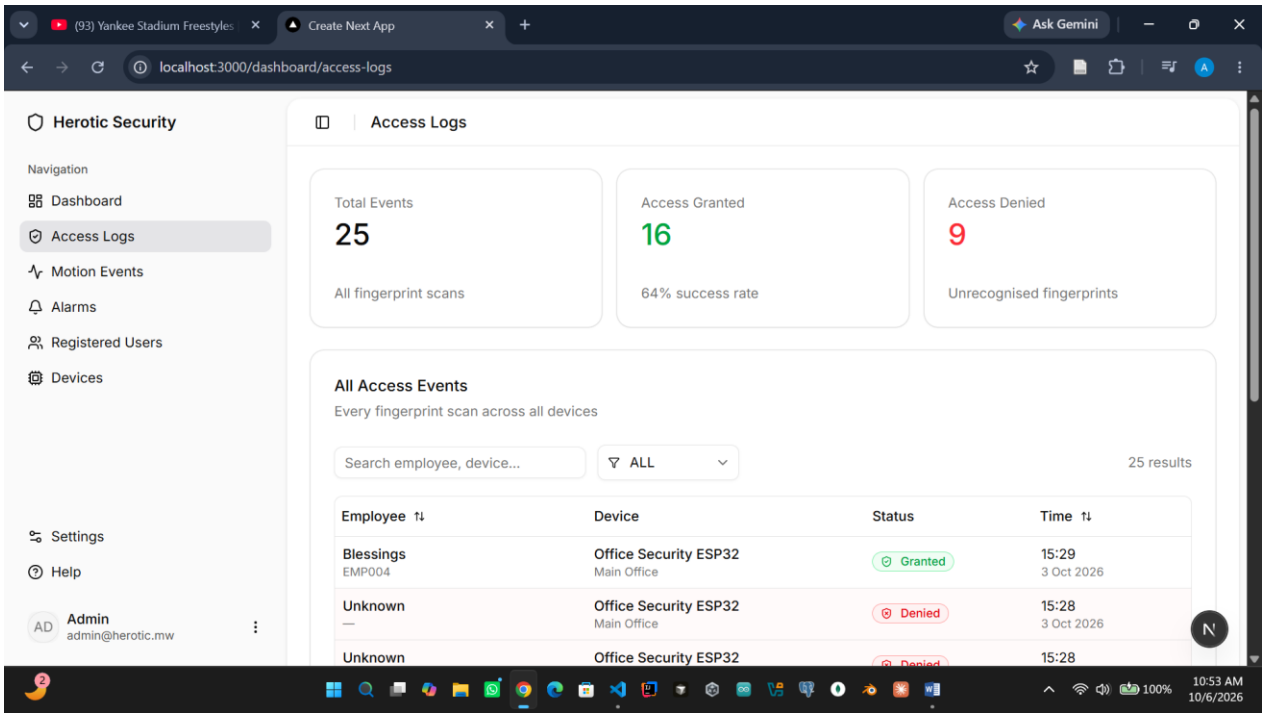

(93) Yankee Stadium Freestyles
Create Next App
Ask Gemini
localhost:3000/dashboard
Herotic Security
No active alarms
Motion and security alerts
0 of 1 ESP32 device connected
IoT security devices
Navigation
Dashboard
Access Logs
Motion Events
Alarms
Registered Users
Devices
Settings
Help
Admin
admin@herotic.mw
Access Events Over Time
Granted vs denied fingerprint scans
Last 3 months
Last 30 days
Last 7 days
Sep 26
Denied 2
Granted 6
Sep 9
Sep 12
Sep 15
Sep 18
Sep 21
Sep 24
Sep 27
Sep 30
Oct 3
Oct 6
Recent Access Logs
All fingerprint scan events
Search employee...
Columns
100%
10:51 AM
10/6/2026


(93) Yankee Stadium Freestyles
Create Next App
Ask Gemini
localhost:3000/dashboard
Herotic Security
Navigation
Dashboard
Access Logs
Motion Events
Alarms
Registered Users
Devices
Settings
Help
Admin
admin@herotic.mw
Recent Access Logs
All fingerprint scan events
Search employee...
Columns
Employee
Device
Status
Time
Blessings
EMP004
Office Security ESP32
Main Office
Granted
03:29 PM
Oct 3
Unknown
Office Security ESP32
Main Office
Denied
03:28 PM
Oct 3
Unknown
Office Security ESP32
Main Office
Denied
03:28 PM
Oct 3
Chembe Mulobwe
EMP-004
Office Security ESP32
Main Office
Granted
03:14 PM
Oct 3
Unknown
Office Security ESP32
Main Office
Denied
03:12 PM
Oct 3
Wathu Msukumbwa
EMP012
Office Security ESP32
Main Office
Granted
03:12 PM
Oct 3
Unknown
Office Security ESP32
Main Office
Granted
04:42 PM
Sep 26
Unknown
Office Security ESP32
Main Office
Denied
04:17 PM
Sep 26
25 total events
Page 1 of 4
100%
10:51 AM
10/6/2026

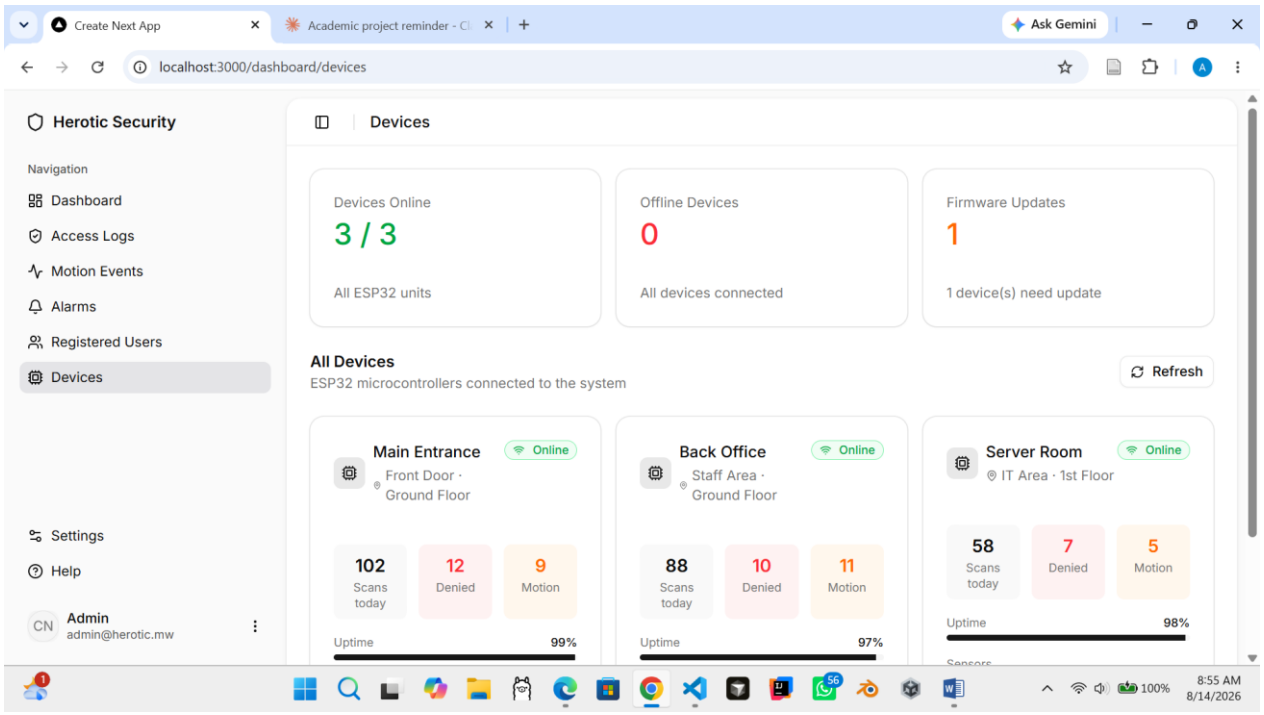
3. User Management Page: This page allows the administrator to enroll, view and deactivate authorized users fingerprint records.

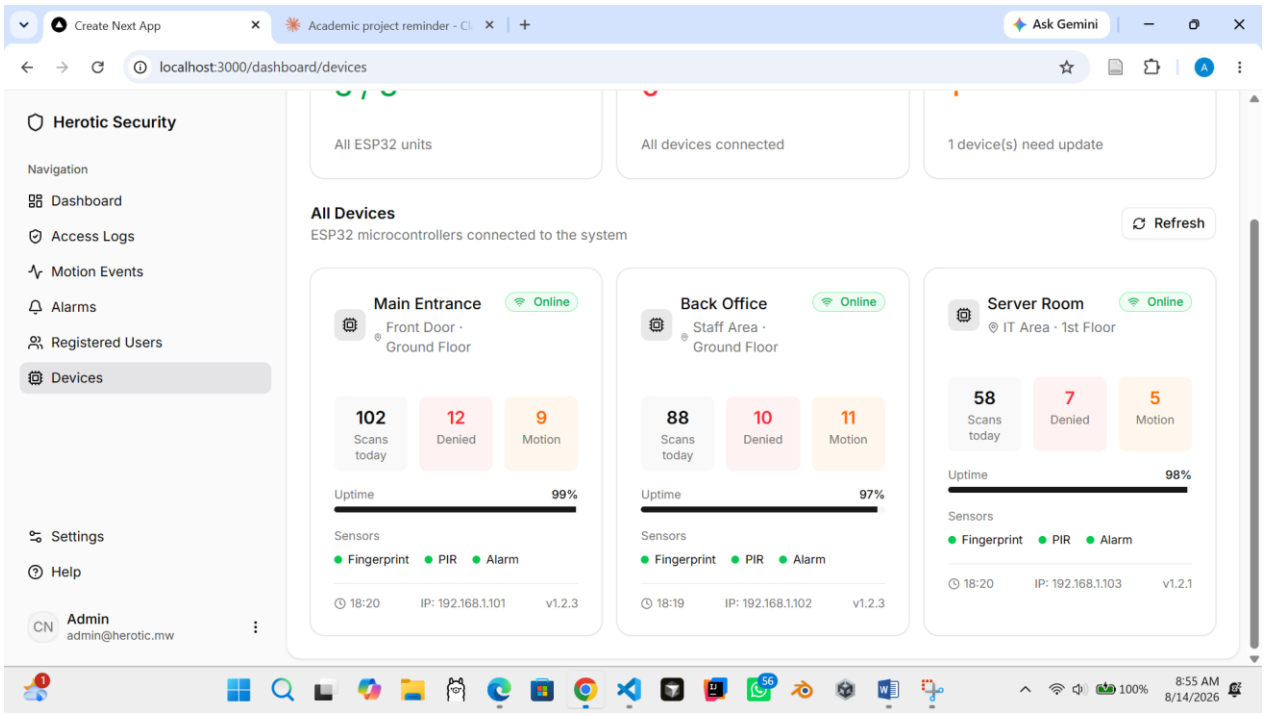


4. Access Log Page: an audit trail of every fingerprint attempt, showing the user, device, time and whether access was granted or denied.



5. Device Management Page: This page shows all registered ESP32 devices with their online/offline status and last-seen timestamps.


Create Next App
Academic project reminder - Cl
Ask Gemini
localhost:3000/dashboard/devices
Herotic Security
Navigation
Dashboard
Access Logs
Motion Events
Alarms
Registered Users
Devices
Settings
Help
Admin
admin@herotic.mw
Devices
Devices Online
3 / 3
All ESP32 units
Offline Devices
0
All devices connected
Firmware Updates
1
1 device(s) need update
All Devices
ESP32 microcontrollers connected to the system
Refresh
Main Entrance
Online
Front Door · Ground Floor
102
Scans today
12
Denied
9
Motion
Uptime
99%
Back Office
Online
Staff Area · Ground Floor
88
Scans today
10
Denied
11
Motion
Uptime
97%
Server Room
Online
IT Area · 1st Floor
58
Scans today
7
Denied
5
Motion
Uptime
98%
100%
8:55 AM
8/14/2026


Create Next App
Academic project reminder - Cl
Ask Gemini
localhost:3000/dashboard/devices
Herotic Security
Navigation
Dashboard
Access Logs
Motion Events
Alarms
Registered Users
Devices
Settings
Help
Admin
admin@herotic.mw
All ESP32 units
All devices connected
1 device(s) need update
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ESP32 microcontrollers connected to the system
Refresh
Main Entrance
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Front Door · Ground Floor
102
Scans today
12
Denied
9
Motion
Uptime
99%
Sensors
Fingerprint
PIR
Alarm
18:20
IP: 192.168.1.101
v1.2.3
Back Office
Online
Staff Area · Ground Floor
88
Scans today
10
Denied
11
Motion
Uptime
97%
Sensors
Fingerprint
PIR
Alarm
18:19
IP: 192.168.1.102
v1.2.3
Server Room
Online
IT Area · 1st Floor
58
Scans today
7
Denied
5
Motion
Uptime
98%
Sensors
Fingerprint
PIR
Alarm
18:20
IP: 192.168.1.103
v1.2.1
100%
8:55 AM
8/14/2026

5.3. Data Design

The data layer's schema is illustrated in Figure 6.5 (Entity Relational Diagram). The real-time security data is modeled across five entities, namely: Devices, FingerprintUsers, AccessLogs, MotionEvents, and AlarmEvents. These are stored and hosted in convex to support reactive, low-latency and real-time updates to and from the dashboard and the database.

Table 5.1: Data Dictionary

Entity	Key Fields	Description
Devices	esp32Id, name, location, status, lastSeen	Registered ESP32 units and their last-known state
FingerprintUsers	fingerprintId, fullName, employeeNumber, department, active	Enrolled staff authorized for biometric access
AccessLogs	deviceId (FK), fingerprintId, status	Every fingerprint scan attempt, granted or denied
MotionsEvents	deviceId (FK), resolved	PIR-triggered motion detections outside working hours
AlarmEvents	motionEventId (FK), deviceId (FK), severity, status	Alerts raised from motion events, tracked to resolution

5.3.1. Normalization

This schema was evaluated against the first three normal forms. All entities satisfy the 1NF, such that they are atomic and have no repeating records. Similarly, the entities satisfy the 2NF, as each table uses a single system generated identifier as its primary key rather than composite keys.

The two deliberate departures from 3NF were retained. First, esp32 is duplicated on *AccessLogs*, *MotionEvents*, and *AlarmEvents* despite being derivable through *deviceId*; this is necessary because the ESP32 firmware authenticates itself by hardware ID and does not hold a reference to its own Convex-assigned *deviceId* at write time. Secondly, the *employeeName* and *employeeNumber* fields are duplicated onto *AccessLogs* rather than resolved through *fingerprintId*; this preserves an accurate audit trail of who accessed the premises or building at the time of the event, even if the corresponding *FingerprintUsers* record is later edited or deactivated.

Both are documented trade-offs of write-path simplicity and audit integrity over strict normalization, rather than oversights.

5.4. Process and Algorithm Design

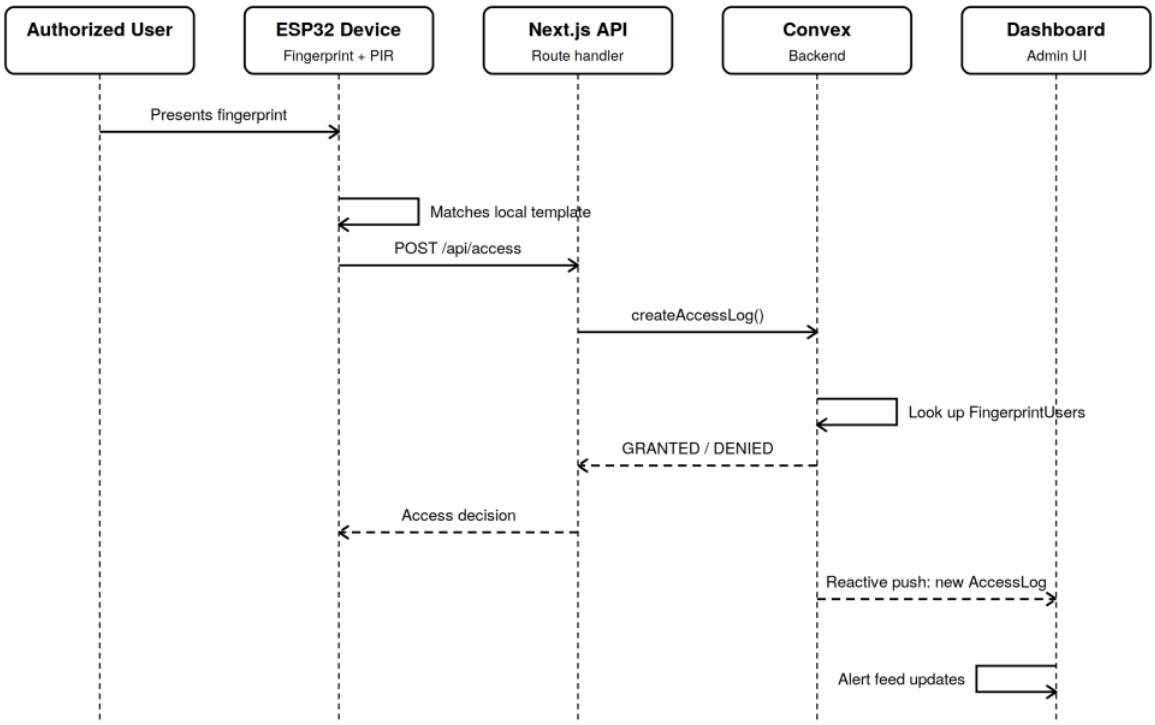


figure 5.1 shows the process and algorithm design.

The sequence of events is as follows:

1. Presentation: The authorized user places a finger on the AS608 sensor.
2. Local matching: The sensor matches the print against its stored templates and returns a fingerprint ID or no match.
3. Request: The ESP32 sends a POST to /api/access with its hardware ID and the fingerprint ID.
4. Decision: The Next.js route calls Convex, which looks up FingerprintUsers. It returns GRANTED for an active match and DENIED otherwise, and writes an AccessLog.
5. Feedback: The ESP32 lights the green or red LED.

6. Update: Convex pushes the new log to the dashboard, which updates without a refresh.

5.5. Hardware Design

The edge device is installed at an entry point of the office. It authenticates fingerprints, sounds an alarm on command, and reports events to the backend over Wi-Fi. The prototype comprises an ESP32 microcontroller, a PIR motion sensor, a fingerprint sensor, a buzzer and an LED. A door motor and an ESP32 camera are planned extensions, described at the end of this section. Figure 5.4 shows the hardware block diagram.

The hardware components and their interfaces are as follows:

1. Central ESP32 Microcontroller
    a. Microcontroller: An ESP32-WROOM-32 module is the central controller of the device. It reads the fingerprint sensor, drives the buzzer and LED, and builds and parses the JSON messages exchanged with the backend. It communicates with the backend through HTTP GET and POST requests over Wi-Fi. It was chosen for its built-in Wi-Fi, low cost and sufficient number of input/output pins.
    b. Bread Board: A 830-point breadboard holds the ESP32 and connects the components through jumper wires. A breadboard was used instead of soldering so that connections could be changed freely during prototyping and the components could be reused.

[Micro controller image]

2. Fingerprint Sensor

a. An AS608 optical fingerprint sensor captures fingerprints and stores the enrolled templates on the sensor itself. When a finger is placed on it, the sensor matches the print against its stored templates and returns the matching ID, or no match. The ESP32 therefore receives only a fingerprint ID and the match result, so raw biometric data is never transmitted over the network. The sensor connects to the ESP32 through a serial (UART) interface on GPIO 16 (RX) and GPIO 17 (TX), and the system uses template IDs 1 to 127.

[Fingerprint sensor image]

3. PIR Motion Sensor
    a. PIR motion sensor: A passive infrared sensor will detect movement outside working hours and send a motion event to the backend, which will raise an alarm and activate the buzzer.

[PIT motion sensor image]

4. Buzzer and LED bulbs
    a. Audible buzzer: A KxG1203C piezo buzzer, connected to GPIO 21, provides an audible deterrent. In the current prototype it is activated manually from the web application, which queues a command that the ESP32 retrieves and executes. Triggering it automatically on motion detection will follow once the PIR sensor is added.
    b. LED bulbs (Red and Green LED bulbs): Two LED bulbs (Red LED Bulb and Green LED bulb) are used to denote the authentication status of a staff member when authenticating. The Green Bulb means authentication is a success, and the Red Bulb means authentication is a fail.

[Buzzer and LED image]

Table 5.2: Hardware components and interfaces

Component	Interface	ESP32 pin	Function
ESP32-WROOM-32	Wi-Fi (HTTP)	n/a	Central controller and backend communication.
AS608 fingerprint sensor	UART	GPIO 16 (RX) and GPIO 17 (TX)	Fingerprint capture, storage and matching
Piezo buzzer	Digital output	GPIO 21	Audible alarm
LED	Digital output	GPIO 4	Authentication indicator
Breadboard and jumper wires	n/a	n/a	Prototype connections

5.5.1. Planned Extensions

The following components were part of the original design but were not available during development. The architecture allows them to be added without changing the backend design.

a. Door motor: A motor will simulate a door opening when authentication succeeds and remain closed when it fails.
b. ESP32 camera: A camera module will capture an image when motion or a failed authentication is detected, so the administrator can verify the event.

5.5.2. Design constraints

The prototype currently covers fingerprint authentication, remote alarm control and device monitoring. Motion-based detection pairs well with visual footage from cameras and AI models to verify unauthorized access. However, due to materials, the project does not include them.

5.6. System Architecture

The proposed system adopts the three-tier architecture consisting of the hardware layer, application layer and data layer as shown in figure 5.1 below. This architecture enables communication between IoT devices and the web application, providing a centralized monitoring and management of security events.

At the hardware layer, the ESP32 microcontroller interfaces with the fingerprint sensor, the PIR sensor and alarm module. The ESP32 is going to process data captured from the sensors and will communicate with the web application through Wi-Fi using HTTP API requests.

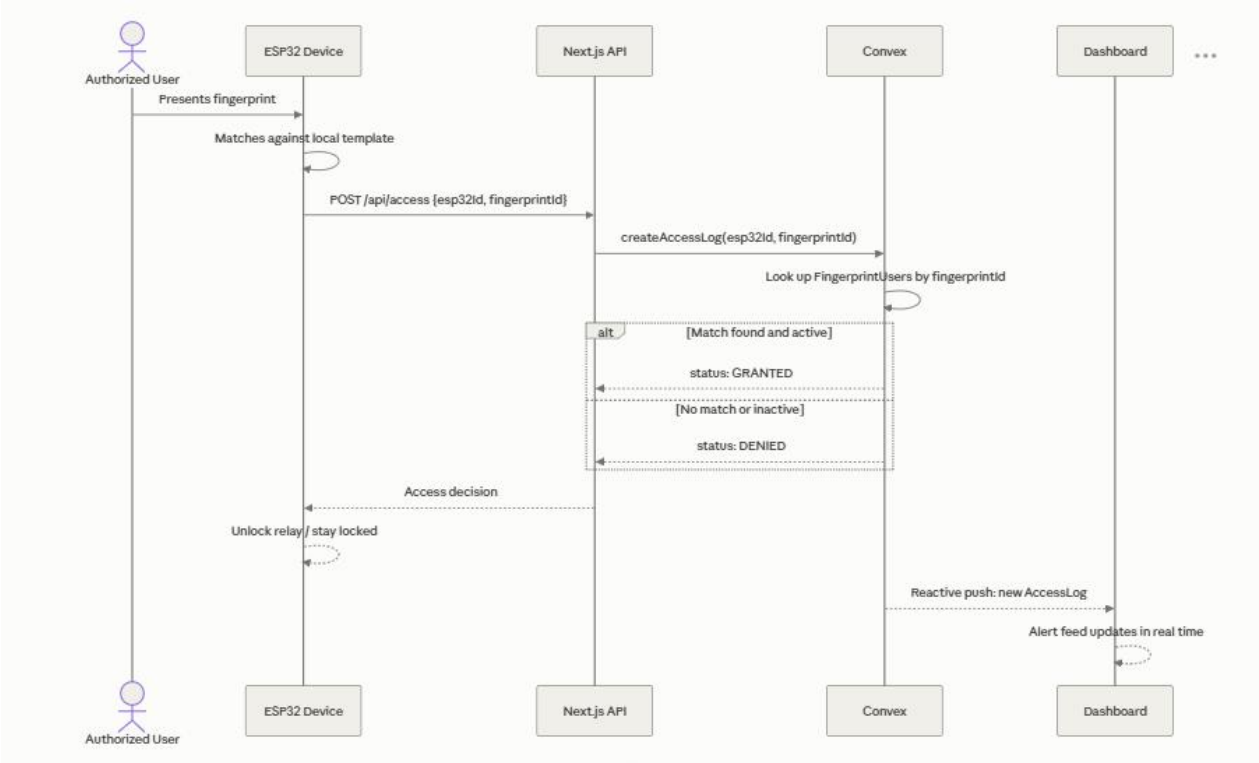
The application layer consists of the web application dashboard developed using Next.js. It will provide administrators will the services to manage authorized users, monitor security events, receive real-time alerts and view security logs. Next.js API routes also serves as the communication interface between the hardware layer (ESP32 microcontroller) and the backend services.

The data layer consists two databases namely: PostgreSQL database and Convex. PostgreSQL permanently stores administrative accounts, manage sessions, and holds authorized user data through roles based access records. Convex on the other hand, handles all real-time services and processes due to its frameworks out-of-the box reactive data access without extensive setup. It handles the storage of fingerprint

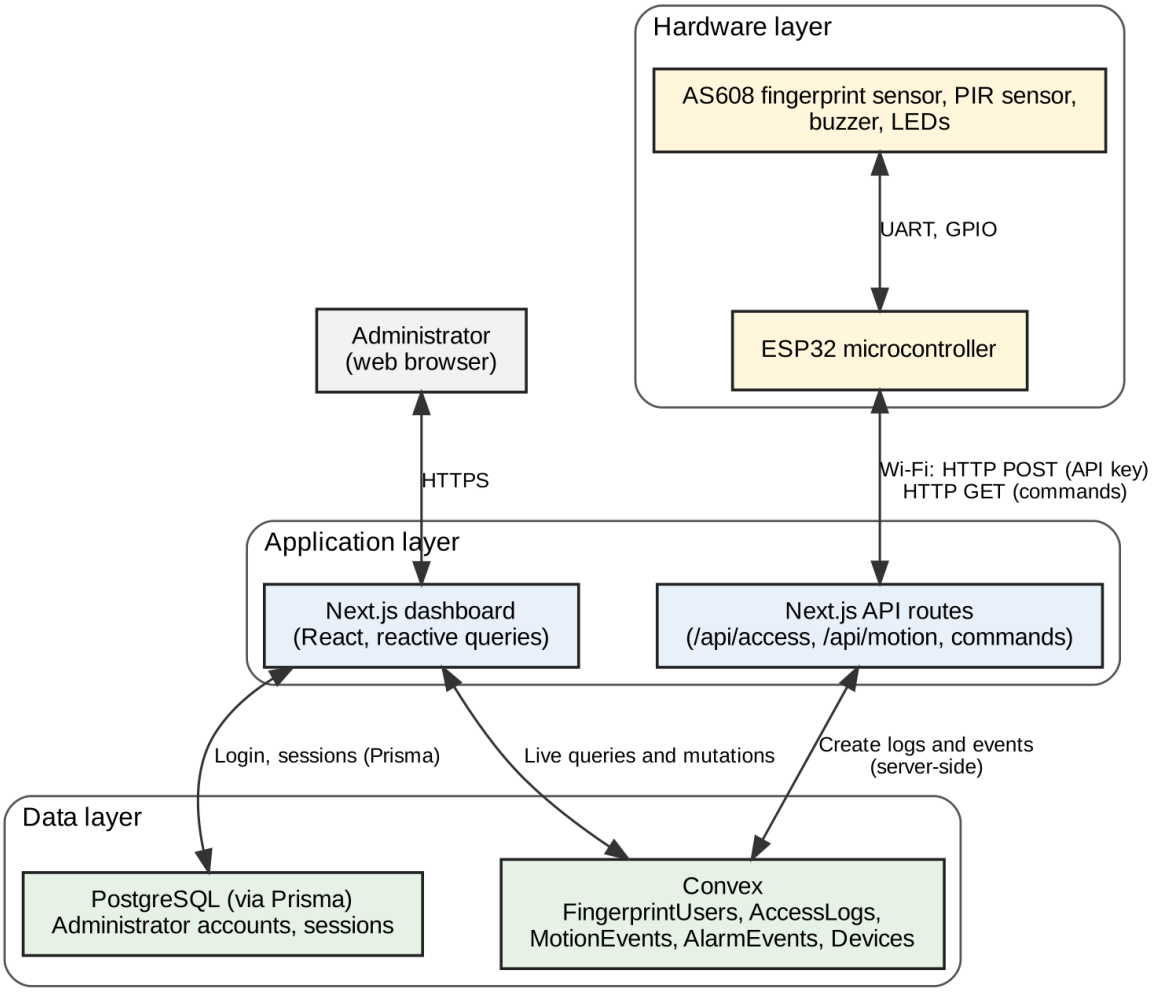
records, access logs, motion detection events, alerts and device information. This layer ensures secure and reliable storage of system data while supporting retrieval by the application.

Additionally, Administrator passwords are hashed using bycrypt, and sessions are protected with session tokens.

This architecture provides a modular, scalable and real-time communication between hardware devices and the web dashboard, making it suitable for SMEs. The figure below illustrated the three-tier architecture discussed in this section.



Figures 5.4 illustrates the communication between the three-layer architecture components.



Figures 5.5 illustrates the system security network including how system components communicate.

Figures 5.4 and 5.5 illustrate the system's two primary event-driven processes. In the authentication flow, the ESP32 performs the fingerprint match locally against the enrolled sensor template before reporting the result to the backend — reducing latency and avoiding transmission of raw biometric data over the network. Convex resolves the fingerprint ID against the FingerprintUsers table to determine grant or deny, and the

decision is pushed reactively to the dashboard without the client polling for updates. The motion-to-alarm flow follows a similar pattern: a PIR trigger outside working hours creates a MotionEvent and, where applicable, an associated AlarmEvent, both surfaced to the Administrator in real time via Convex's reactive queries. The Administrator's acknowledgment is the only point where a human interaction re-enters the flow, closing the loop from detection to resolution.

CHAPTER 6: PROTOTYPING AND TESTING

6.1. Introduction

This chapter describes how the system design in Chapter 5 was turned into a working prototype. Two streams of work ran in parallel: hardware development of the physical edge prototype device, and software development of the web-based monitoring system. The software follows the Model-View-Controller (MVC) pattern (Pressman & Maxim, 2019). The chapter ends with system testing, which checks that the two streams work together as one system.

6.2. Hardware Development: The Physical Prototype

The prototype is a single edge device placed at the entry and exit point of a simulated office, as designed in Section 5.5 (Figure 6-1).

[Insert Figure 6-1: Integrated Physical Prototype, a photo of the whole device.]

6.2.1. Assembly

The components were first assembled on a breadboard so that wiring could be corrected without soldering. This suited the prototyping approach in Section 3.4, where the circuit was changed several times after testing. The assembly (Figure 6-2) was done in this order:

1. The ESP32-WROOM-32 was seated on the 830-point breadboard.
2. The AS608 fingerprint sensor was connected to the ESP32's UART (GPIO 16 as RX, GPIO 17 as TX) and powered from the board's supply rails.
3. The KxG1203C piezo buzzer was connected to GPIO 21.
4. The authentication LEDs were connected to GPIO 4 through resistors. Green shows access granted and red shows access denied.

5. [CONFIRM: The PIR motion sensor was connected with its output on GPIO __ and its VCC and GND on the shared rails. If the PIR is not wired yet, say so here and move it to 6.2.3.]

[Insert Figure 6-2: Wiring schematic or breadboard photo, and Figure 6-3: Fingerprint sensor close-up.]

6.2.2. Firmware

The ESP32 firmware was written in **Arduino C++ using Arduino IDE version 2.3.10** and uploaded through USB. It does the following:

1. It connects to Wi-Fi and retries if the connection drops.
2. It reads the fingerprint sensor. The sensor matches the scan against its stored templates (IDs 1 to 127), so only the fingerprint ID and match result reach the ESP32. Raw biometric data never leaves the sensor.
3. It sends the result to the backend as a JSON message in an HTTP POST, with the device's API key.
4. It polls the backend for queued commands, such as a buzzer command from the dashboard, and carries them out.
5. It sends a periodic heartbeat so the dashboard can show the device as online or offline.
6. On a PIR trigger outside working hours, it sends a motion event and sounds the buzzer.

6.2.3. Hardware Limitations

The door motor and the ESP32 camera were not available during development. They remain planned extensions as stated in Section 5.5.1. The backend design does not need to change to add them.

The ESP32 limitation on using ESP-camera was catered by using the camera on a mobile phone to simulate visual monitoring.

6.3. Software Development: The MVC Framework

The software followed the Model-View-Controller pattern.

6.3.1. View

The View is the Next.js dashboard that the administrator uses. It subscribes to Convex queries, so new events appear on screen without a page refresh. The view contains the following routes:

1. **Login Page** (Figure 6-4): The only public page. It asks for administrator credentials before any content is shown.

2. **Main Dashboard** (Figure 6-5): Shows device status, the latest alerts and summary statistics, plus a small log of the day's biometric activity.

3. **User Management Page** (Figure 6-6): Lets the administrator enroll, view and deactivate fingerprint users.

4. **Access Log Page** (Figure 6-7): an audit trail of every fingerprint attempt, granted or denied.

5. **Device Management Page** (Figure 6-8): lists each registered ESP32 with its online or offline status and last-seen time.

[Insert your five screenshots here, one per figure, each with its caption.]

6.3.2. Model

The Model is the database structure, built on the Convex backend. The Entity-Relationship Diagram in Section 5.3 shows how the tables relate. Tables 6-1 to 6-5 list each table and its fields.

Table 6.1: Devices Table

Field	Type	Description
esp32Id	String	Hardware ID the firmware uses to identify itself
name	String	Descriptive name of the device
location	String	Where the device is installed
status	String	Online or Offline
lastSeen	Number	Time of the last heartbeat. This is achieved by converting number to time, using new Date(lastSeen).

		The other reason is because convex does not support Date datatype.

Table 6.2: FingerprintUsers Table

Field	Type	Description
fingerprintId	number	Template ID stored on the sensor (1 to 127)
fullName	String	Name of the enrolled staff
employeeNumber	String	Staff number
Department	String	Department of the staff
Active	Boolean	Whether the person may currently gain access

Table 6.3: AccessLogs Table

Field	Type	Description
deviceId	Id(Foreign Key -> Devices)	Device that took the scan.
fingerprintId	Number	ID returned by the sensor.
employeeName, employeeNumber	String	Copied at the time of the event to preserve the audit trail.
Status	String	Granted or Denied

Table 6.4: MotionEvents Table

Field	Type	Description
deviceId	Id(Foreign Key -> Devices)	Device whose PIR sensor triggered
Resolved	Boolean	Whether the administrator has dealt with the event

Table 6.5: AlarmEvents Table

Field	Type	Description
motionEventId	Id (Foreign Key)	The motion event that raised the alarm.
deviceId	Id (Foreign Key -> Devices)	Device concerned.
Severity	String	Severity level of the alarm.
Status	String	Progress of the alarm to resolution.

PostgreSQL, accessed through Prisma, stores administrator accounts and sessions. The reasons for duplicating some fields on the log tables are explained in Section 5.3.1.

6.3.3. Controller

The Controller is the back-end logic and firmware that processes data, applies the rules, and connects the Model, the View and the hardware. Five functions are described below, tied to the functional requirements in Table 4.1. Their code is in Appendix C.

1. **Fingerprint authentication (FR3, FR4), firmware.** [function name, e.g.

checkFingerprint()] reads the sensor, takes the matched ID or a no-match result, and sends it to the backend. (Appendix C, Snippet C.1).

2. **Access decision (FR3, FR4), Convex.** The access mutation looks up the fingerprint ID in FingerprintUsers. It returns GRANTED if an active user is found and DENIED otherwise, then writes an AccessLog entry. (Snippet C.2)
3. **Motion alert and alarm (FR5), Convex.** When a motion event arrives, the backend records it and creates an AlarmEvent for the dashboard. (Snippet C.3)
4. **Real-time dashboard data (FR6, FR8).** A Convex query returns the latest logs, alerts and device statuses. The React useQuery hook subscribes to it, so the dashboard updates the moment data changes. A scheduled check marks devices offline when heartbeats stop. (Snippet C.4)
5. **Access control for administrators (FR1).** Middleware **[name your auth method, e.g. session check]** blocks every dashboard route unless the user has a valid administrator session, and redirects everyone else to the login page. API routes used by the ESP32 are protected by the device API key. (Snippet C.5).

6.4. System Testing

Testing checks that the system works as expected and meets the requirements. Evidence-based testing was used: every test has a stated purpose, steps, expected output, and proof of the result in the form of a screenshot, serial log or dashboard record (Myers et al., 2011). Fingerprint tests used consenting participants in a controlled environment (Section 3.8).

6.4.1. Test Scenarios

Table 6.6: Functional Requirement Test Case

Test ID	Description	Correct data/steps	Pass output	Wrong data/steps	Fail output
T1	Administrator login (FR1)	Enter valid credentials	Dashboard opens	Enter a wrong password	Access refused, error shown
T2	Enroll a user (FR2, FR7)	Enter name, number and a free fingerprint ID.	User saved and listed	Reuse an existing ID	Entry rejected
T3	Authorized access (FR3)	Enrolled finger on sensor	GRANTED, green LED, log entry	Finger scanned with the device offline	Request fails, error handled by firmware
T4	Unauthorized access (FR4)	Unenrolled finger	DENIED, red LED, denial logged	Deactivated user scans	No alarm raised
T5	Motion alert (FR5)	Movement outside working hours	Motion event and alarm appear on dashboard	Movement during working hours	No alarm raised
T6	Real-time alerts (FR6)	Trigger an event with the	New entry appears	Dashboard connection lost	Dashboard stops updating

		dashboard open	without refresh		
T7	Device status (FR8)	Power the device off	Device shown as offline	Device powered on again	Device shown as online again
T8	Device authentication	POST with correct API key	Request accepted	POST with no or wrong key	Request rejected

6.4.2. Test Parameters and Evidence Based result

These are the measured parameters from Objectives 4, with what was done for each.

Table 6.7: Testing Parameters

Table 6.8: Evidence-Based Testing

6.4.3. Error Handling and Fixes

Three problems were found during testing and corrected before the final test run:

1. Unreliable alert delivery.
   - Symptom: Some fingerprint scans were not logged on the dashboard.
   - Cause: The serial monitor showed the ESP32's Wi-Fi connection timing out before the HTTP POST finished.
   - Fix: The firmware was changed to retry the connection and resend the request when it fails, with a timeout of 5 seconds.

2. Fingerprint read failures.
    - Symptom: Valid fingers were sometimes rejected.
    - Cause: Poor placement, and sometime dry or dirty fingers, produced low-quality images on the AS608 optical sensor.
    - Fix: Enrollment was repeated with captures per finger, and the LED feedback was changed so the user knows when to place the finger again.
3. Device identity mismatch
    - Symptom: The backend rejected log entries from the device.
    - Cause: The ESP32 firmware identifies itself by hardware ID and does not hold the Convex-assigned device ID when it writes an event.
    - Fix: The hardware ID is stored on the log tables and resolved on the server.

6.5. Summary of Results

Testing showed that the prototype works as designed. (8 of 8) functional tests (T1 to T8) passed. The evaluation against the testing parameters gave a false accept rate of 0%, a false reject rate of 0%, an average motion-to-dashboard alert time of 3 seconds, and 100% of events delivered during the reliability run. Each objective was met:

- the hardware module was developed
- the web dashboard was developed
- the two were integrated through the HTTP API
- the prototype was evaluated against the defined parameters

The prototype hardware cost MWK 225,000, against 2,500,000 for comparable commercial products, which answers the cost-based research question. For full test evidence, refer to Appendix D: Evidence-Based Testing Screenshots.

CHAPTER 7: CONCLUSION

7.1. Conclusion

This project set out to address the weaknesses of traditional security in small and medium-sized enterprises (SMEs) in Blantyre. Key-based access can be duplicated, manual surveillance cannot verify who enters, and incidents are often found only after damage has occurred. The study confirmed through interviews and observation that many SMEs rely on these methods because commercial systems from vendors such as ZKTeco, Bosch and Hikvision are costly and complex to set up and maintain.

In response, a functional prototype of a biometric and motion-based IoT security system was developed and tested. The prototype achieves the following:

1. **Biometric access control.** An AS608 fingerprint sensor, connected to an ESP32 microcontroller, verifies users locally. Only the fingerprint ID and the match result are sent over the network, so raw biometric data is never transmitted.
2. **Motion detection and alarms.** A PIR sensor detects movement outside working hours, raises an alarm event on the dashboard and sounds the buzzer.
3. **Centralized real-time monitoring.** A Next.js dashboard backed by Convex shows access attempts, motion events, alarms and device status as they happen, and lets the administrator manage authorized users.
4. **Affordability.** The prototype was built from low-cost, widely available components. The hardware cost was MWK 225,000, compared with **MWK 2,500,000** for comparable commercial products.

Against the objectives set in Chapter 1, the hardware module was developed, the web-based dashboard was developed, the two were integrated through a RESTful HTTP API, and the prototype was evaluated against defined parameters. The main research

question, on how such a system can be built at a lower cost than comparable products, is therefore answered by the prototype itself and by the cost comparison.

7.2. Limitations of the Study

The study had the following limitations:

1. **Controlled testing environment.** The prototype was tested in a controlled setting and not in a live business. It has not yet faced real network instability, daily heavy use or varied lighting and temperature.
2. **Hardware constraints.** The low-cost components, including the ESP32 and the PIR sensor, limit detection range and accuracy. This trade-off was accepted to keep the system affordable.
3. **Dependence on internet connectivity.** Real-time monitoring and the dashboard depend on a stable connection. During an outage, events are delayed.
4. **Components not available.** The door motor and the ESP32 camera were not available during development, so physical door control and visual verification of alerts are not part of the current prototype.
5. **Limited participants.** Requirements were gathered from SMEs in Blantyre only, and fingerprint testing used a small group of consenting participants, so the accuracy figures cannot be generalised to all users.

7.3. Personal Reflection

Building this system showed me that an IoT project is only as strong as its weakest link. During testing, some fingerprint scans were not logged on the dashboard. The serial monitor showed that the ESP32's Wi-Fi connection was timing out before the HTTP

request finished. I changed the firmware to retry the connection and resend the request, with a 5-second timeout, and the scans were then logged reliably.

This taught me that in a security system, designing for failure matters as much as designing the detection itself. A device that cannot report an event, or that reports it late, fails at its main job.

The project also showed me how much of the work lies in integration. Each part, such as the sensor, the firmware, the API and the dashboard, worked alone before it worked together, and the real effort was in making them agree on data formats, device identity and timing.

7.4. Future Work

This prototype is a base for a production system. The following are recommended:

1. **Add the door motor and an ESP32 camera.** A motor would control the door itself, and a camera would capture an image on a failed authentication or motion event, so the administrator can confirm what happened.
2. **Develop a mobile Application and Add notifications.** Develop a mobile application and Implement SMS or push alerts allowing real time alerts for administrators who are away from the dashboard.
3. **Add local caching on the ESP32.** Events and access attempts would be stored during an internet outage and sent once the connection returns.
4. **Support several devices and more sensors.** The architecture already separates devices from the backend, so more entry points and sensors, such as door contacts, can be added without redesign.
5. **Improve biometric security.** Add a second factor, such as a PIN or card, for sensitive areas, and extend the audit trail with reports for the owner.

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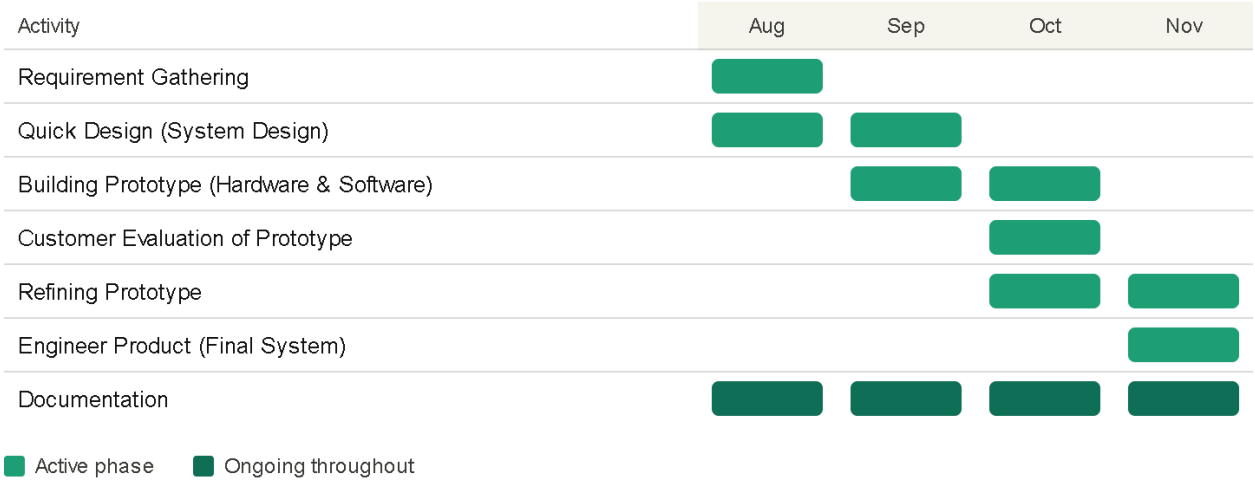
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Appendix A: Project Work Plan

The project will be conducted over a period of four months starting from August to November of 2026. Activities will begin with requirement gathering and system design, followed by parallel hardware and software development. System integration and testing will then be carried out, followed by evaluation. Documentation will be done throughout the project.



Appendix B: Project Budget

Item	Description	Cost (MWK)
ESP32 Microcontroller	Main IoT Controller	70, 000
Fingerprint Sensor	Biometric authentication	50, 000
PIR motion sensor	Motion detection	20, 000

Alarm/Buzzer	Alarm system	25, 000
Connecting components (Wires and Breadboard)	Connectors and cables	30, 000
Relay switch	Connecting to mains	30, 000
Internet Bundle	Development and Testing	80, 000
Laptop Maintenance	Development resources	100, 000
Stationery	Printed papers and writing materials	15, 000
Web hosting	Website storage	80, 000
Miscellaneous	Contingency cost	50, 000
Total		550,000